

Theory of Computation

Exercise 1: (Mathematic preliminary, Language, String)

1. Let $\Sigma = \{a, b\}$ and $L = \{aa, bb\}$. Describe $\bar{L}$ by a set notation.

$$\bar{L} = \{a^n b^m; n \geq 3, m \geq 3\} \cup \{w \in \Sigma^*; |w| \geq 3\}$$

$$\cup \{\lambda, a, b, ab, ba\}$$

2. Find five strings which are in each of the following languages.

a) $L = \{w \in \{a\}^*: |w| \bmod 3 \neq |w| \bmod 2\}$

`{ aa, aaa, aaaaa, aaaaaaaaa, aaaaaaaaa }`

b) $L = \{w \in \{a, b\}^*: n_a(w) \geq n_b(w) + 1\}$

Where $n_a(w)$ means the number of a's in string w.

`{ a, aa, aaa, aab, aba, baa }`